

Deep Learning 1

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$$\text{OKP} (0.3 \times \text{TK3} + \underbrace{0.3 \text{BA3} + 0.25 \text{MA3} + 0.15 \text{TP}}_{\text{Итого} \geq 7.5})$$

$$\text{Итого} \geq 7.5$$

- Backpropagation
- Fully-connected networks
- ? Classification

$$a(x) = w_1 x_1 + \dots + w_d x_d = f(x, w) \quad \begin{array}{l} x \in \mathbb{R}^d \\ w \in W \text{ weights} \end{array}$$

Input features

$$\{(x_i, y_i)\}_{i=1}^l \quad \text{training set}$$

$$L(\hat{y}, y) - \text{loss function}$$

$$Q(w) = \frac{1}{l} \sum_{i=1}^l L(f(x_i, w), y_i) \rightarrow \min_w$$

Backprop. alternatives

Historic:

- Evolutionary algorithm 1950s
- Reservoir computing (2000) $\rightarrow \text{NN} \rightarrow \text{lin reg}$

Modern:

- Direct feedback alignment (2016)

$$y = f(g(x)) \quad \frac{dy}{dx}(x) = \left. \frac{df}{dg} \right|_{g(x)} \cdot \left. \frac{dg}{dx} \right|_x \quad x \leftarrow \text{fixed random}$$

- Forward-forward (2022)

Backpropagation

$$1. \quad y = f(g(x)) \quad \frac{dy}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}$$

$$2. \quad y = f(g_1(x), g_2(x), \dots, g_n(x))$$

$$\frac{dy}{dx} = \sum_{i=1}^n D_i f \frac{dg_i}{dx}$$

$$D_i f = \frac{df}{dz_i} \quad f(z_1 \dots z_n)$$

symbolic $x^2 \rightarrow \underbrace{2}_x \underbrace{x}_x \quad \forall x$

numeric $x^2 \rightarrow \{x := 3\} \rightarrow 6$

Computational Graph

$$f = x_1^2 (x_2 x_3 - 2x_1^2) = x_1^2 x_2 x_3 - 2x_1^4$$

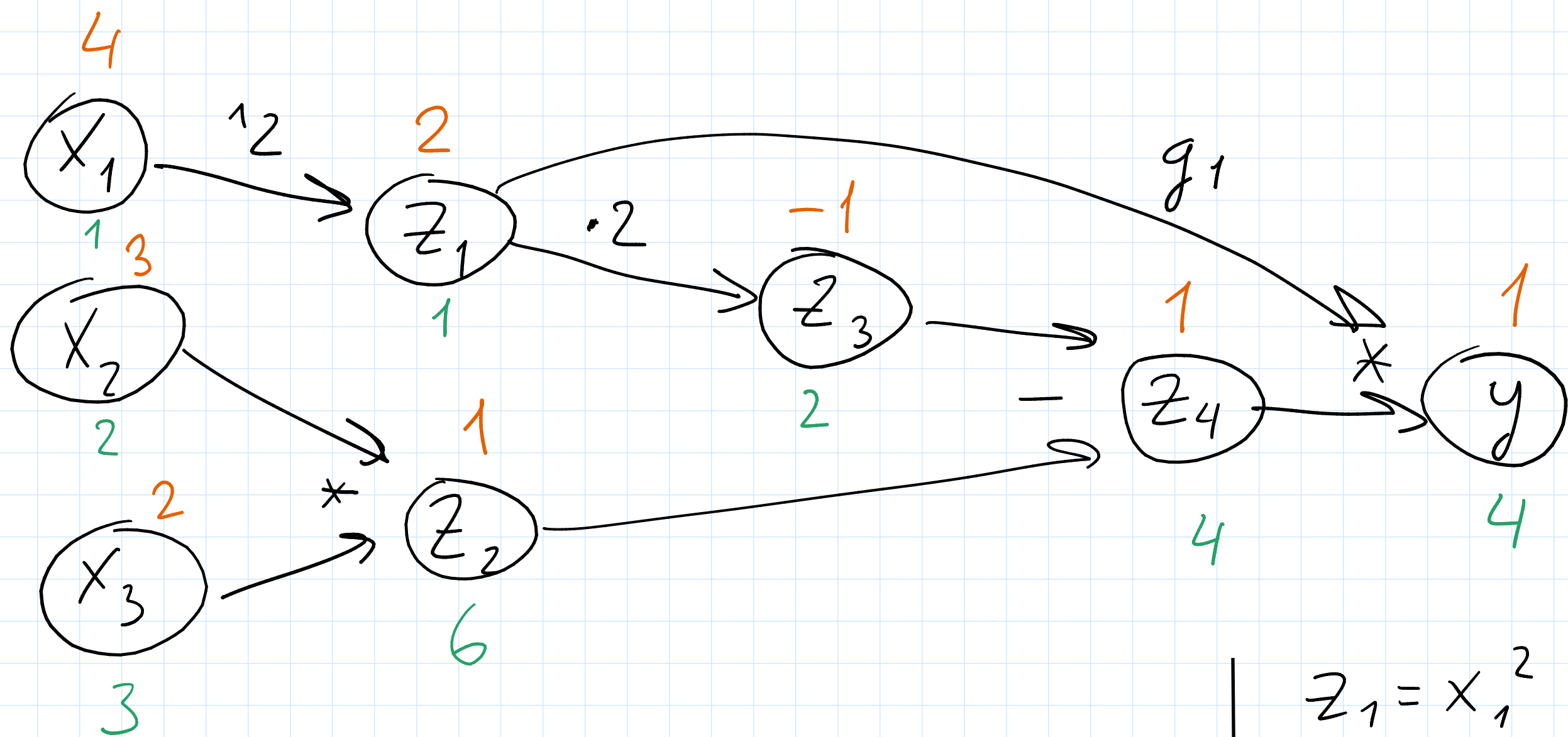
$$\frac{df}{dx_1} = 2x_1 x_2 x_3 - 8x_1^3 = 4$$

$$\frac{df}{dx_2} = x_1^2 x_3 = 3$$

$$\frac{df}{dx_3} = x_1^2 x_2 = 2$$

$$x = (1, 2, 3)$$

$$y = x_1^2 (x_2 x_3 - 2x_1^2)$$



$$\begin{aligned}
 z_1 &= x_1^2 \\
 z_2 &= x_2 x_3 \\
 z_3 &= 2z_1 \\
 z_4 &= z_2 - z_3 \\
 y &= z_1 z_4
 \end{aligned}$$

$$\frac{dy}{dy} = 1$$

$$\frac{dy}{dz_4} = \frac{dy}{dy} \cdot \frac{dy}{dz_4} = 1 \cdot z_1 = 1$$

$$\frac{dy}{dz_3} = \frac{dy}{dz_4} \cdot \frac{dz_4}{dz_3} = 1 \cdot (-1) = -1$$

$$\frac{dy}{dz_1} = z_4 \cdot 1 + \frac{dy}{dz_3} \cdot \frac{dz_3}{dz_1} = 4 + (-1) \cdot 2 = 2$$

$$y = f(g_1(z_1), g_2(z_1)) = g_1 g_2 = z_1 z_4$$

$$g_1(z_1) = z_1 \quad g_2(z_1) = z_4(z_1)$$

$$\frac{dg_1}{dz_1} = 1$$

$$\frac{dg_2}{dz_1} = \frac{dz_4}{dz_1} = \frac{dz_4}{dz_3} \cdot \frac{dz_3}{dz_1}$$

$$\frac{dy}{dz_2} = \frac{dy}{dz_4} \cdot \frac{dz_4}{dz_2} = 1 \cdot 1 = 1$$

$$\frac{dy}{dx_1} = \frac{dy}{dz_1} \cdot \frac{dz_1}{dx_1} = 2 \cdot 2x_1 = 4$$

$$\frac{dy}{dx_2} = \frac{dy}{dz_2} \cdot \frac{dz_2}{dx_2} = 1 \cdot x_3 = 3$$

$$\frac{dy}{dx_3} = \frac{dy}{dz_2} \cdot \frac{dz_2}{dx_3} = 1 \cdot x_2 = 2$$

$$x \in \mathbb{R}^d \quad a(x) = \langle w, x \rangle = w^T x \quad w \in \mathbb{R}^d$$

$$z_1 = W_1 x + b_1 \quad W_1 \in \mathbb{R}^{d_1 \times d} \quad b_1 \in \mathbb{R}^{d_1}$$

$$z_2 = W_2 z_1 + b_2 \quad W_2 \in \mathbb{R}^{d_2 \times d_1} \quad b_2 \in \mathbb{R}^{d_2}$$

...

$$z_n = W_n z_{n-1} + b_n \quad W_n \in \mathbb{R}^{d_n \times d_{n-1}} \quad b_n \in \mathbb{R}^{d_n}$$

$$z_n = \underbrace{W_n \dots W_2 W_1}_{W \in \mathbb{R}^{d_n \times d}} x + b$$

$\sigma(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ activation function (non-linear)

$$z_i = \sigma(W_i z_{i-1} + b_i)$$

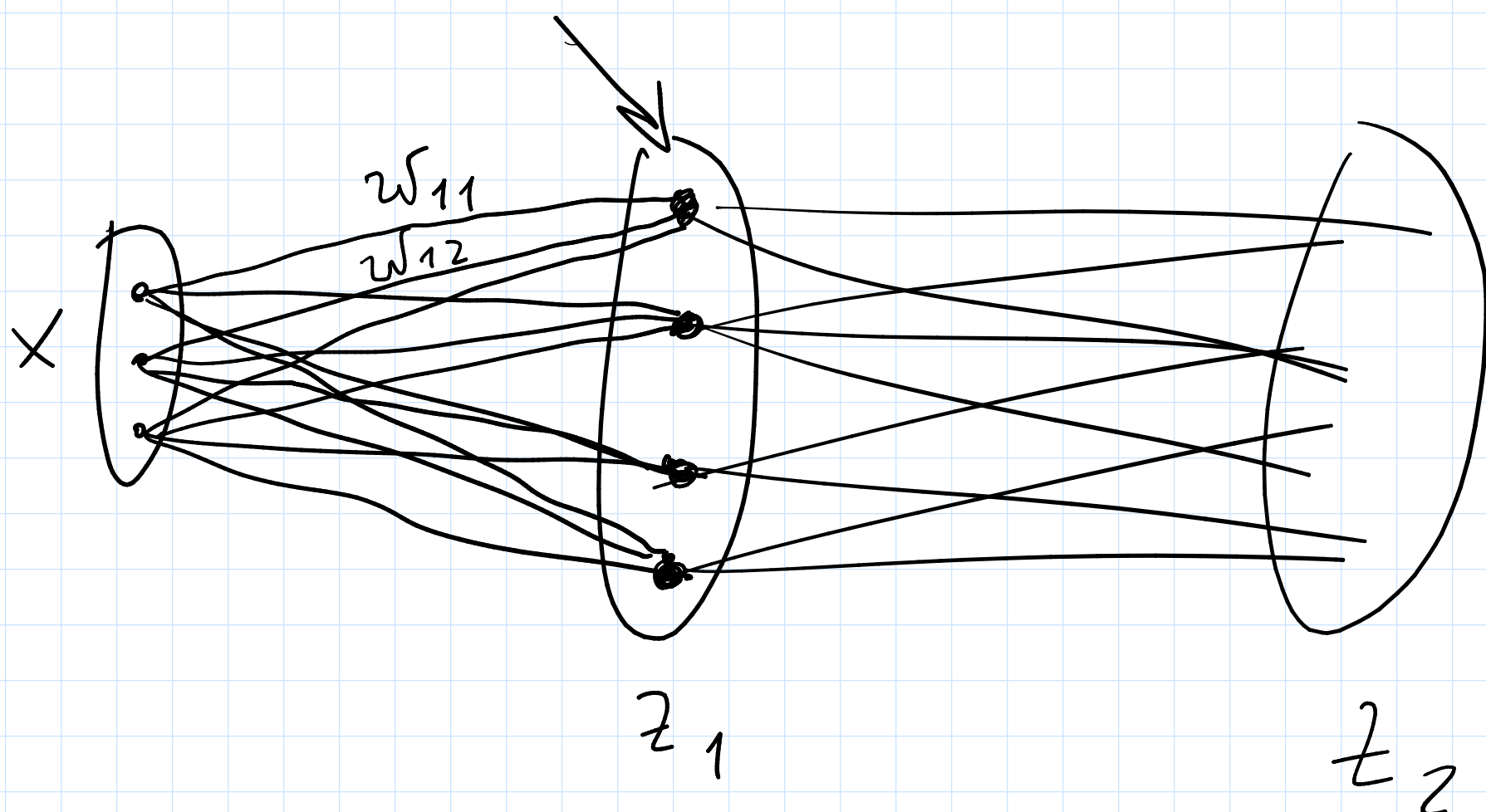
dense / linear / fully-connected layer

z_i - latent representation

$$z_i = (z_i^{(1)}, \dots, z_i^{(d_i)})$$

↑

neuron



MLP Multi Layer Perceptron

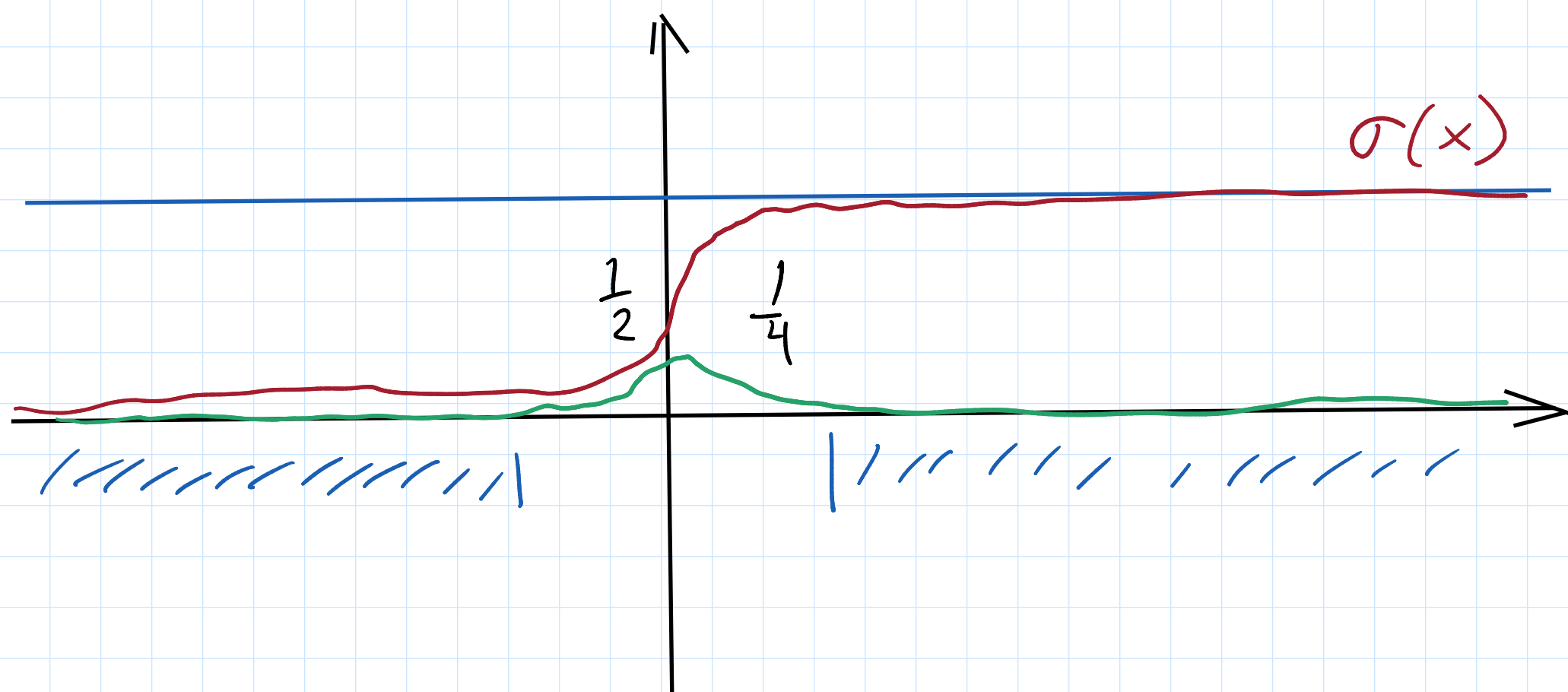
FCN Fully Connected Network

1. Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(1-\sigma)$$

$$\frac{e^{-x}}{(1+e^{-x})^2}$$



$$u_i = W_i z_i + b_i$$

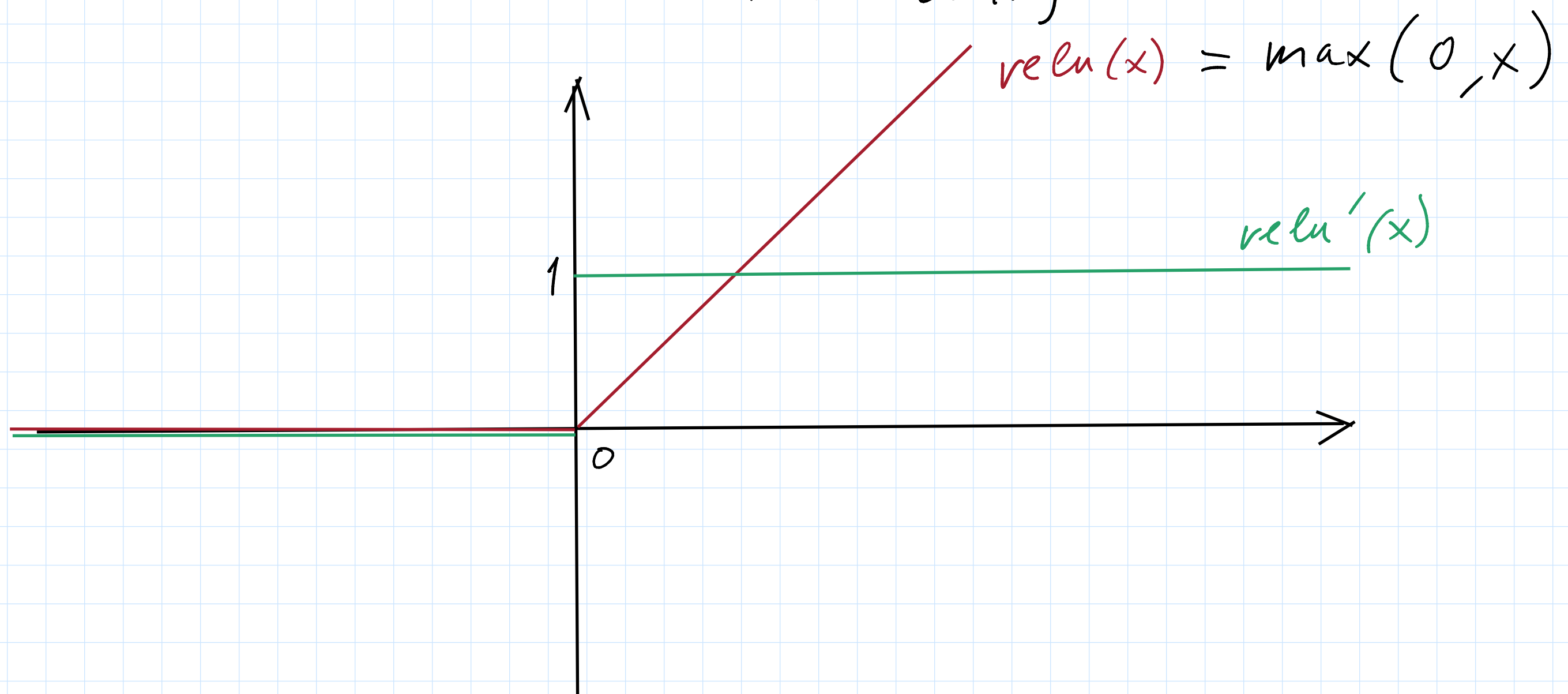
$$z_{i+1} = \sigma(u_i)$$

$$\frac{d\ell}{dz_i} = \frac{d\ell}{du_{i+1}} \cdot \underbrace{\sigma'(u_i)}_{\approx 0} \cdot \frac{du_{i+1}}{dz_i}$$

≈ 0

vanishing gradients

2. ReLU (rectified linear unit)



3. Leaky ReLU (α) = $\max(\alpha x, x)$

